MATHEMATICS

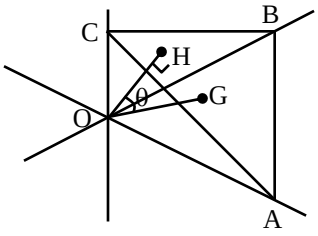
SECTION – I

Single Correct Choice Type

1. **(b)** $\Delta^2 = \Delta_x^2 + \Delta_y^2 + \Delta_z^2$

$$= \left(\frac{1}{2}\frac{p^2}{lm}\right)^2 + \left(\frac{1}{2}\frac{p^2}{mn}\right)^2 + \left(\frac{1}{2}\frac{p^2}{ln}\right)^2$$

$$\Rightarrow \Delta = \frac{p^2}{2lmn}$$



Also $\left(\frac{l^2+m^2+n^2}{3}\right)^3 \geq l^2m^2n^2 \quad \because A.M. \geq G.M.$

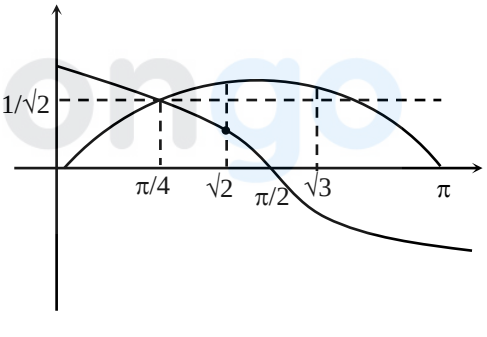
$$\Rightarrow \frac{p^4}{4(lmn)^2} \geq \frac{27p^4}{4} \Rightarrow \Delta \geq \frac{3\sqrt{3}p^2}{2}$$

2. **(c)** The number of common vertices are the number of common roots to the equation $Z^{1982} - 1 = 0$ and $Z^{2973} - 1 = 0$, which is GCD of 1982, 2973, i.e., 991

3. **(a)** As $\sqrt{2} < \frac{\pi}{2} < \sqrt{3} \Rightarrow \sin\sqrt{2} > 0$ and

$\cos\sqrt{3} < 0$

$\Rightarrow \sin\sqrt{2} - \cos\sqrt{3} > 0$



Also $\sqrt{3} < \frac{3\pi}{4} \Rightarrow \sin\sqrt{3} > \frac{1}{\sqrt{2}}$

$\sqrt{2} > \frac{\pi}{4} \Rightarrow \cos\sqrt{2} < \frac{1}{\sqrt{2}}$

$\Rightarrow -\cos\sqrt{2} + \sin\sqrt{3} > 0$

$\therefore$ Curve is an ellipse of form $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$= \left|\sin\sqrt{2} - \cos\sqrt{3}\right| > \left|\sin\sqrt{3} - \cos\sqrt{2}\right|$

4. **(d)** $\frac{r}{4r^4+1} = \frac{r}{(2r^2-2r+1)(2r^2+2r+1)}$

$$= \frac{1}{4}\left[\frac{(2r^2+2r+1)-(2r^2-2r+1)}{(2r^2-2r+1)(2r^2+2r+1)}\right]$$

$$= \frac{1}{4}\left[\frac{1}{(2r^2-2r+1)} - \frac{1}{(2r^2+2r+1)}\right]$$

$$\Rightarrow \sum_{r=1}^{n} \frac{r}{4r^4+1} = \frac{1}{4}\left[1 - \frac{1}{2n^2+2n+1}\right] = \frac{\sum n}{4\sum n + 1}$$

5. **(c)** $f(x)=e^{x-k}$ then inverse of $f(x)$; $f^{-1}(x)=(\ln x)+k$,
and also both functions are increasing, therefore
$f(x)=f^{-1}(x)$ is equivalent to $f(x)=f^{-1}(x)=x$
$\Rightarrow \quad \ln x+k=x$ should have a solution
$\Rightarrow \quad k=x-\ln x$
Now, let $g(x)=x-\ln x$
has least value 1 as $g'(x)=1-\dfrac{1}{x}$ has a minimum at $x=1$
and $\lim\limits_{x\to 0^+} g(x)$, $\lim\limits_{x\to\infty} g(x)$ both approach to ∞.

6. **(a)** $(A-B)C=BA^{-1}$
$\Rightarrow \quad AC-BC-BA^{-1}+AA^{-1}=I_3$
$\Rightarrow (A-B)(C+A^{-1})=I_3 \quad \Rightarrow \quad (A-B)^{-1}=C+A^{-1}$
$\Rightarrow (C+A^{-1})(A-B)=I_3 \quad \Rightarrow \quad C(A-B)+A^{-1}A-A^{-1}B=I_3$
$\Rightarrow C(A-B)=A^{-1}B$

7. **(c)** Given every element of set *B* has been mapped to some element of set *A*.
Without loss of generality we can assume
$b_1<b_2<b_3.....<b_n$
Consider a_1, a_2, ...a_{2n} be divided into *n* sub-groups, so that each b_i is mapped to one sub group and all elements mapped to a particular b_i to hold equality.
Therefore the correct answer is the number of ways in which $2n$ elements can be divided into *n* non-empty sub groups i.e., ${}^{2n-1}C_n$.

8. **(c)** Let $x_n=\tan\theta_n$

Now, $\tan\theta_{n+1}=x_{n+1}=\dfrac{x_n}{1+\sqrt{1+x_n^2}}=\dfrac{\tan\theta_n}{1+\sqrt{1+\tan^2\theta_n}}$

$\Rightarrow \quad \tan\theta_{n+1}=\dfrac{\tan\theta_n}{1+\sec\theta_n}=\dfrac{\sin\theta_n}{1+\cos\theta_n}=\tan\dfrac{\theta_n}{2}$

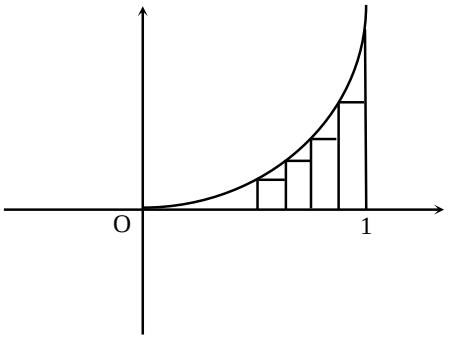
$\Rightarrow \quad \theta_{n+1}=\dfrac{\theta_n}{2} \Rightarrow \theta_n=\dfrac{\theta_{n-1}}{2}$

Now, $\theta_1=\dfrac{\pi}{3} \Rightarrow \theta_n=\dfrac{\pi}{3.2^{n-1}} \Rightarrow x_n=\tan\left(\dfrac{2\pi}{3.2^n}\right)$

$\Rightarrow \quad \lim\limits_{n\to\infty} 2^n x_n=\lim\limits_{n\to\infty}\dfrac{\tan(2\pi/3.2^n)}{(1/2^n)}=\dfrac{2\pi}{3}$

SECTION – II
Multiple Correct Choice Type

9. (a), (c) & (d)

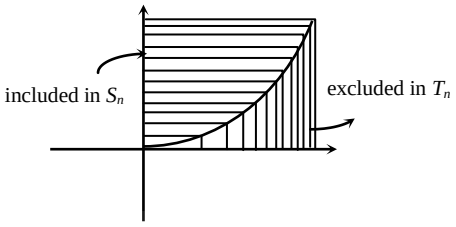


$$S_n = \lim_{n\to\infty}\left(\frac{1}{n^6}+\frac{32}{n^6}+\frac{243}{n^6}+.....\frac{1}{n}\right)$$

$$T_n = \lim_{n\to\infty}\left(\frac{1}{n^6}+\frac{32}{n^6}+\frac{243}{n^6}+.....\frac{(n-1)^5}{n^6}\right)$$

$$S_n = \lim_{n\to\infty}\sum_{r=1}^{n}\frac{r^5}{n^6} = \int_0^1 x^5\ dx = \frac{x^6}{6}\Big|_0^1 = \frac{1}{6}$$

$$\Rightarrow\quad S_n = \left(\frac{1}{6}\right)^+;\ T_n = \left(\frac{1}{6}\right)^-$$



But since x^5 is concave upward the area included in S_n for any two consecutive values of r is more than area excluded in T_n for same values of r.

$$\Rightarrow\quad S_n - \frac{1}{6} > \frac{1}{6} - T_n \quad\Rightarrow\quad S_n + T_n > \frac{1}{3}$$

10. (b), (c)

Let $S = \sum_{j=1}^{n}\sum_{k=1}^{n}\frac{z_j}{z_k} = (z_1 + z_2 + ...z_n)\left(\frac{1}{z_1}+\frac{1}{z_2}+.....\frac{1}{z_n}\right)$

If $|z_1| = |z_2| .....|z_n| = r$

$$\Rightarrow\quad \frac{1}{z_k} = \frac{\bar{z}_k}{r^2}$$

$$\Rightarrow\quad S = \frac{1}{r^2}(z_1 + z_2 + ...z_n)(\bar{z}_1 + \bar{z}_2 + ...\bar{z}_n)$$

$$= \frac{|z_1 + z_2 + ...z_n|^2}{r^2}$$

But $z_1 + z_2 + ...z_n = |z| = r$

$$\Rightarrow\quad S = \frac{r^2}{r^2} = 1$$

11. **(b)** $(f'(x))^2 - k^2(f(x))^2 \le 0$

$\Rightarrow \quad (f'(x) - kf(x))\,(f'(x) + kf(x)) \le 0$

$\Rightarrow \quad \left(f(x)e^{-kx}\right)' \left(f(x)e^{kx}\right)' \le 0$

$\Rightarrow$ Exactly one of the functions $g_1(x) = f(x)e^{-kx}$ or $g_2(x) = f(x)e^{kx}$ is non decreasing.

But $f(0) = 0 \Rightarrow$ both functions g_1 and g_2 have a value zero at $x = 0$

$\forall x \in [0, 1]$, $g_1(0) = 0$ and g_1 increasing $\Rightarrow \; g_1(x) \ge 0 \;\Rightarrow\; f(x) \ge 0$

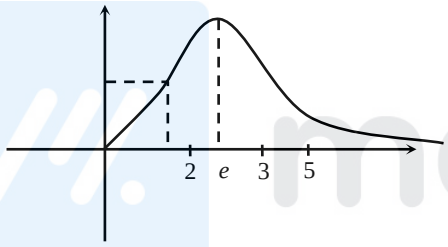
$g_2(0) = 0$ and g_2 decreasing $\Rightarrow \; g_2(x) \le 0 \;\Rightarrow\; f(x) \le 0$

$\Rightarrow \quad f(x) = 0 \; \forall x \in [0, 1]$

12. **(b), (c) & (d)**

Solving we get $x = 2^{1/2}$, $y = 3^{1/3}$, $z = 5^{1/5}$

Using graph of $x^{1/x}$



$\Rightarrow \quad 3^{1/3} > 5^{1/5}$

Also $2^{1/2} < 3^{1/3}$	$2^{1/2} > 5^{1/5}$
as $2^3 < 3^2$	$2^5 > 5^2$

$\Rightarrow \quad y > x > z$

Hence (b), (c) & (d) are NOT true.

SECTION–III
Comprehension Type

13. **(c)** Normal at point P is angle bisector of angle SPS' (which is also obvious as angle bisector of $\angle BPC$ is common tangent to E_1 and E_2).

Since angle bisector of AC and BD are perpendicular E_3 is orthogonal to E_1 and E_2 at P.

For E_3, $2ae = BC$ and $2a = BP + PC$

$\Rightarrow \quad e = \dfrac{BC}{BP + PC}$

Moreover if circumcircle of ΔBPC would have touched ellipse, angle bisector of BPC would have been normal to circle which will be true if BPC is isosceles.

14. **(b)** $e_1 = \dfrac{AB}{PA+PB}$; $e_2 = \dfrac{CD}{PC+PD}$ and $e_3 = \dfrac{BC}{PB+PC}$

Since ΔPAB and ΔPDC are similar $\dfrac{PB}{PC} = \dfrac{PA}{PD} = \dfrac{AB}{CD} = \lambda$

$\Rightarrow \; e_1 = e_2$

15. **(a)** $MB = \dfrac{AB(PB)}{PB+PA} = PBe_1$

$CN = PCe_2 = PCe_1$

Adding $MB + CN = e_1(PB+PC) = \dfrac{e_1 BC}{e_3} \quad \Rightarrow \quad \dfrac{MB+CN}{BC} = \dfrac{e_1}{e_3}$

SECTION–IV
Matrix–Match Type

1. **A-2** $f''(0) = \dfrac{1+0}{1+f(0)} = \dfrac{1}{2} \;\Rightarrow\; f'(x)$ is increasing at $x = 0$

$\Rightarrow$ for some $\delta > 0$ $f'(x) > 0$, $\forall\, x \in (0, \delta)$

$\Rightarrow$ $f(\delta) > f(0) \;\Rightarrow\; f(\delta) > 1$ …(i)

$f''(\delta) = \dfrac{1+\delta}{1+f(\delta)} > 0 \;\Rightarrow\; f'(x)$ is increasing at $x = \delta \Rightarrow\; f'(\delta) > 0$

Hence $f(x)$ is increasing at $x = \delta$

Repeat the same argument on interval right to $x = \delta$.

It comes out $f'(x) > 0$, $\forall x > 0$.

Hence $f(x)$ is increasing $\forall x > 0$.

B-2 $f(x) = e^x$ which is increasing function.

C-2, 4 If $f(x)$ is polynomial function of degree n, then

$h(x) = f(x) - x$ is also a polynomial of degree n

Now, $f(x^2+1) = f^2(x) + 1$

$x = 0 \;\Rightarrow\; f(1) - 1 = 0 \;\Rightarrow\; h(1) = 0$

$x = 1 \;\Rightarrow\; f(2) - 2 = 0 \;\Rightarrow\; h(2) = 0$

……..

……

$x = r \;\Rightarrow\; f(r) - r = 0 \;\Rightarrow\; h(r) = 0$ for $r \in N$

Keep on repeating it until $r > n$

$\Rightarrow$ $h(x) = 0$ has more than n roots

Hence $h(x) = 0$, $\forall\; x \in R$ (it's an identity)

$\Rightarrow$ $f(x) = x$, $\forall\, x \in R$ i.e., linear and increasing.

D-2, 4 From given functional equation it is obvious that $f(x)$ has to be a linear polynomial.

Let $f(x)=ax+b$

$\Rightarrow \quad f(ax+b)=6x-(ax+b) \quad \Rightarrow \quad a(ax+b)+b=6x-ax-b$

$\Rightarrow \quad x(a^2-b+a)=0$ and $ab+2b=0$

Solving these equation, $a=2$ and $b=0$ or $a=-3$ and $b=0$

$\Rightarrow \quad f(x)=2x$ or $f(x)=-3x$

But $f(x)=-3x$ is not $R^+ \to R^+$

Hence $f(x)=2x$ i.e., linear and increasing.

2. **(A)–2; (B)–1; (C)–1; (D)–3**

Total number of triangles equals to ${}^8C_3=56$

Total number of equilateral triangles whose one vertex is A equals to 3. ($\Delta AFH, \Delta AFC$ and ΔAHC). Now total number of vertices equals to 8 and all the triangle made by three vertices, so total number of equilateral triangles are equal to $\frac{8\times 3}{3}=8$

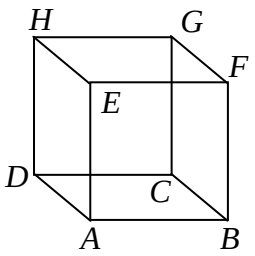
Total number of isosceles triangle in plane $ABCD$ is 4.

$\Rightarrow$ Total number of isosceles triangles (not equilateral) are equal to 4 × 6 = 24

$\Rightarrow$ Now total number of scalene triangles $={}^8C_3-24-8=24$

Now, all proper isosceles triangle (not equilateral) are right angled triangles. Because scalene triangle is formed by one edge, one force diagonal and one cube diagonal, so all scalene triangle are right angle triangled.

$\Rightarrow$ Total right angled triangle = 24 + 24 = 48



SECTION–V
Integer Answer Type

1. Given $g(x_1)g(x_2)g(x_3)g(x_4)g(x_5)=A$

$=(x_1^2-2)(x_2^2-2)(x_3^2-2)(x_4^2-2)(x_5^2-2)$

$=-(2-x_1^2)(2-x_2^2)(2-x_3^2)(2-x_4^2)(2-x_5^2)$ ..(i)

$=-[2^5-(\sum x_1^2)2^4+\sum x_1^2.x_2^2.2^3-\sum x_1^2.x_2^2.x_3^2.2^2$

$+\sum x_1^2.x_2^2.x_3^2.x_4^2.2-x_1^2.x_2^2\ x_3^2.x_4^2.x_5^2\ ]$

$p(x)=x^5+x^2+1=0$ has roots $x_1,\ x_2,\ ....,x_5$, then that equation $q(x)$ whose roots are square of the roots of $p(x)$ is $q(x)=(\sqrt{y})^5+(\sqrt{y})^2+1=0$; $\alpha=x$ and $y=\alpha^2$

$\Rightarrow \quad (y+1)^2 = \left(-\sqrt{y}\right)^{5\times 2}$

$\Rightarrow \quad y^2+2y+1=y^5 \quad \Rightarrow \quad q(x)=y^5-y^2-2y-1=0,$

then $\sum x_1^2 = \sum y_1 = 0$

$\sum x_1^2.x_2^2 = \sum y_1.y_2 = 0$

$\sum x_1^2.x_2^2.x_3^2 = \sum y_1.y_2.y_3 = 1$

$\sum x_1^2.x_2^2.x_3^2.x_4^2 = \sum y_1.y_2.y_3.y_4 = -2$

$\sum x_1^2.x_2^2.x_3^2.x_4^2.x_5^2 = \sum y_1.y_2.y_3.y_4.y_5 = 1$, then

$A = -[2^5 - 0 + 0 - 2^2 - 2.2 - 1] = -[32-4-4-1] = -[32-9]$

$= -23$

$x_1x_2x_3x_4x_5 = -1 \Rightarrow g(x_1x_2.....x_5) = -1$

$\Rightarrow g(x_1)g(x_2).....g(x_5) - 30g(x_1x_2.....x_5) = 7$

Ans. 7

Alternative:

Let us form that equation having roots $y = g(x_i)$ i.e., $y = x^2 - 2$

$x = \sqrt{y+2}$

$\Rightarrow \quad \left(\sqrt{y+2}\right)^5 + \left(\sqrt{y+2}\right)^2 + 1 = 0$

$\Rightarrow \quad y^5 + 20y^4 + 40y^3 + 79y^2 + 74y + 23 = 0$

$\therefore \quad g(x_1).....g(x_5)$ = Product of roots

$= -23$

$x_1x_2x_3x_4x_5 = -1 \Rightarrow g(x_1x_2.....x_5) = -1$

$\Rightarrow g(x_1)g(x_2).....g(x_5) - 30g(x_1x_2.....x_5) = 7$

Ans. 7

2 . $f(x)+f(\alpha x)+f(\alpha^2 x)+.....+f(\alpha^6 x) = 7A_0 + \sum_{k=1}^{20} A_k x^k \left(1+\alpha^k+.....+\alpha^{6k}\right)$

but when $k \neq 7$ and $k \neq 14$, then $1+\alpha^k+\alpha^{2k}+....+\alpha^{6k} = 0$

Hence $f(x)+f(\alpha x)+.....+f(\alpha^6 x) = 7A_0 + 7A_7x^7 + 7A_{14}x^{14}$

$= 7\left(A_0 + A_7x^7 + A_{14}x^{14}\right)$

$k = 7$

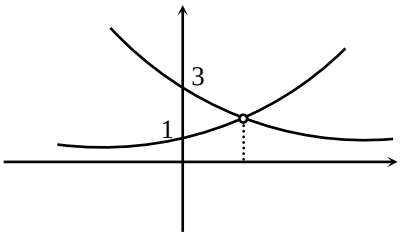
Ans. 7

3. Let $f^{-1}(x)$ be $g \Rightarrow \ln\left(g+\sqrt{g^2+1}\right) = x$

$\Rightarrow \quad g+\sqrt{g^2+1} = e^x \qquad \ldots(i)$

and $-g+\sqrt{g^2+1} = e^{-x} \qquad \ldots.(ii)$

Now, $\left|e^x - e^{-x}\right| = 2e^{-|x|}$



Case I: $x > 0$; $e^{-|x|} = e^{-x}$ and $e^{x} > e^{-x}$

$\Rightarrow \; e^{x} - e^{-x} = 2e^{-x} \;\Rightarrow\; e^{x} = 3e^{-x}$

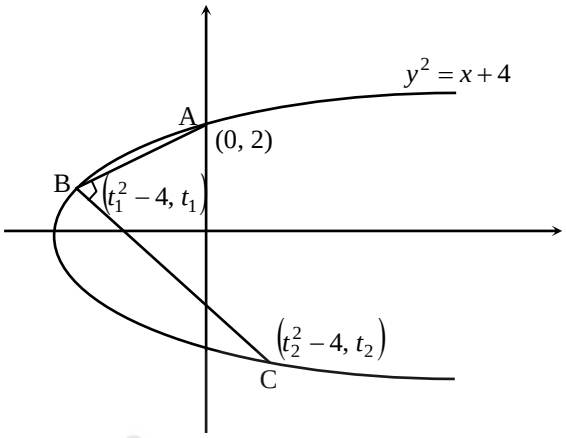
Case II: $x < 0$; $e^{-|x|} = e^{x}$ and $e^{x} < e^{-x}$

$\Rightarrow \; e^{-x} - e^{x} = 2e^{x} \;\Rightarrow\; e^{-x} = 3e^{x}$

$\therefore$ Two solutions.

Ans. 2

4. Now $\left(\frac{t_2 - t_1}{t_2^2 - t_1^2}\right)\left(\frac{t_1 - 2}{t_1^2 - 4}\right) = -1$



put $t_1 \neq t_2$ and $t_1 \neq 2$

$\Rightarrow \; \frac{1}{(t_1 + t_2)}\frac{1}{(t_2 + 2)} = -1$

$\Rightarrow \; t_1^2 + t_1 t_2 + 2t_1 + 2t_2 + 1 = 0$

Quadratic in t_1 should have real roots.

$\Rightarrow \; (t_2 + 2)^2 - 4(2t_2 + 1) \geq 0 \;\Rightarrow\; t_2^2 - 4t_2 \geq 0$

$\Rightarrow \; t_2 \leftarrow (-\infty, 0] \cup [4, \infty)$

There least positive value of t_2 is 4.

Ans. 4